



Reasoning (from Dieter Fensel's lecture, Pascal Hierzl's lecture, my editing)

Knowledge Base and Reasoning

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- Inferring new knowledge from existing knowledge based on formal logical rules. In the context of an OWL (Web Ontology Language)-based ontology
 - Check ontology consistency
 - Discover implicit class hierarchies
 - Determine individual membership in a class
 - Discover logical redundancies or contradictions in semantic models



- Utilize Description logics
- Description logics are a family of knowledge representation languages that are used to represent the knowledge of an application domain in a structured and formally well-understood way.
- The tableau algorithm attempts to build a model for a concept by decomposing it using a set of expansion rules. If all possible attempts **lead to a contradiction**, the concept is deemed unsatisfiable



- Reasoner
 - Parsing Ontologi.
 - Konsistensi Description Logic (DL) sebagai bahasa formal dari Ontology
 - Scan semantik secara keseluruhan
 - Run logic algorithm for inferencing
 - Detect problem
 - Deduktive reasoning → for finding new axioms from existing axioms



- Protege has included a few reasoner
 - ELK
 - FaCT++
 - HermiT → OWL 2 DL.
 - Mendukung keseluruhan SROIQ(D)
 - Pellet → Ontology/OWL with Semantic Web Rule Language
 - dll



- Ex: Ontology Family Relationship

Class hierarchy: owl:Thing

Annotations: owl:Thing

Annotations +

Description: owl:Thing

Equivalent To +

SubClass Of +

General class axioms +

SubClass Of (Anonymous Ancestor)

Instances +

Target for Key +

Disjoint With +

Disjoint Union Of +

Ontology metrics:

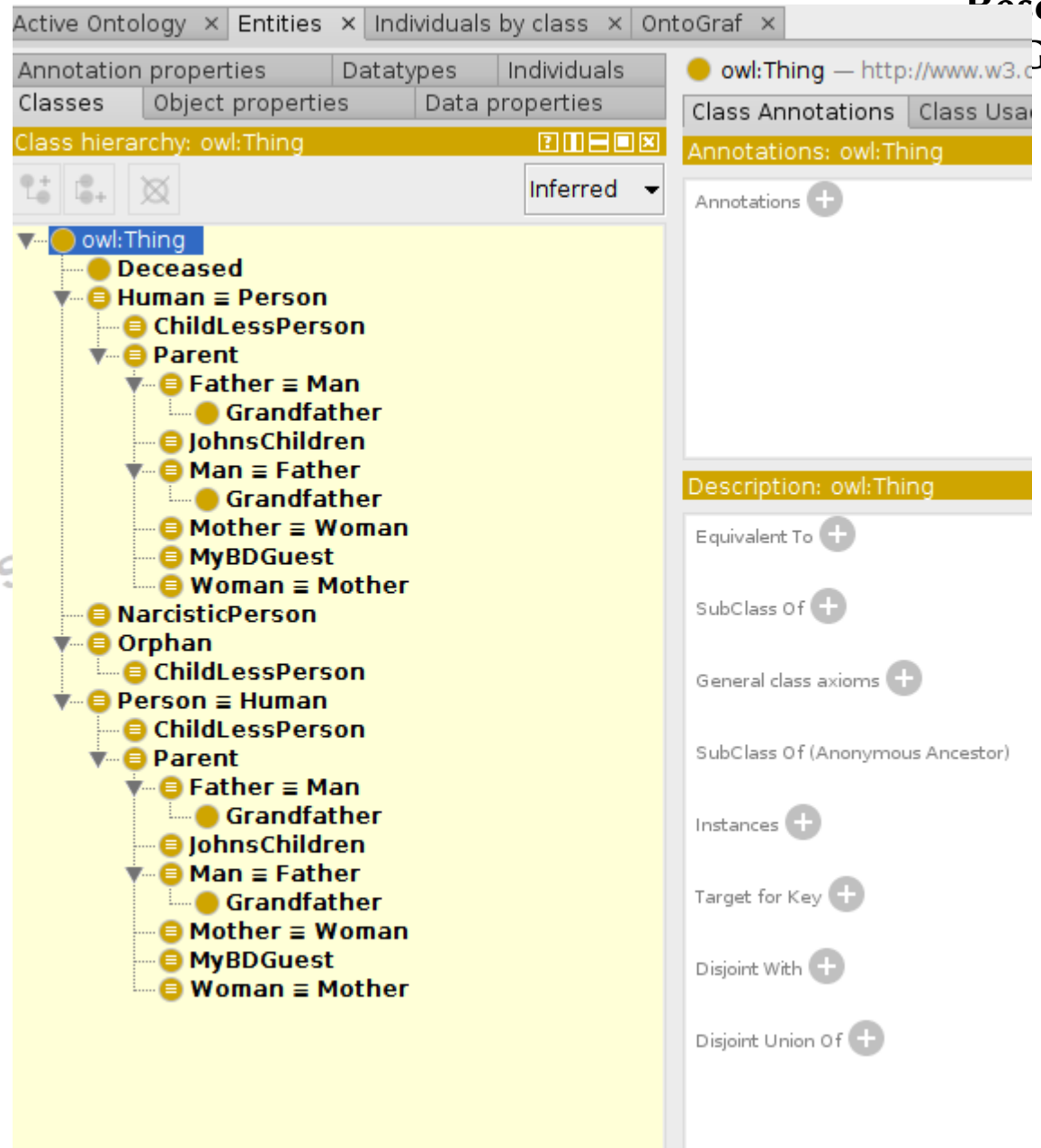
Metrics	
Axiom	102
Logical axiom count	58
Declaration axioms count	37
Class count	15
Object property count	9
Data property count	5
Individual count	8
Annotation Property count	3
DL expressivity	SROIQ(D)

Class axioms

- owl:Thing
 - Deceased
 - Human $\equiv$ Person
 - JohnsChildren
 - MyBDGuest
 - NarcisticPerson
 - Orphan
 - Parent
 - Father $\equiv$ Man
 - Grandfather
 - Mother $\equiv$ Woman
 - Person $\equiv$ Human
 - ChildLessPerson
 - Man $\equiv$ Father
 - Grandfather
 - Woman $\equiv$ Mother



- After reasoning





Ex: ...Detect problem

- Class: Mother, Class: Person, Class: Child
- Child SubClassOf Person
- ObjectProperty: hasChild, ObjectProperty: hasMother
- hasChild Domain Mother, hasChild Range Child → karakter Functional dan Inverse Functional
- hasChild InverseOf hasMother
- Individual: Mary, Individual: Minni, Individual: Missy
- Mary hasChild Missy, Minni hasMother Mary
- Minni DifferentFrom Missy

Explanation for: owl:Thing SubClassOf owl:Nothing

Minni hasMother Mary

hasChild InverseOf hasMother

Functional: hasChild

Mary hasChild Missy

Minni DifferentFrom Missy



Tableau Logic Algorithm

Ex:

- $\text{Mother} \sqsubseteq \top$, $\text{Person} \sqsubseteq \top$, $\text{Child} \sqsubseteq \top$,
- $\text{Child} \sqsubseteq \text{Person}$
- $\text{Domain}(\text{hasChild}) = \text{Mother} \rightarrow \exists \text{hasChild}.\top \sqsubseteq \text{Mother}$
- $\text{Range}(\text{hasChild}) = \text{Child} \rightarrow \top \sqsubseteq \forall \text{hasChild}.\text{Child}$
- $\text{Func}(\text{hasChild})$, $\text{InvFunc}(\text{hasChild})$
- $\text{Mary hasChild Missy} \rightarrow \text{hasChild}(\text{Mary}, \text{Missy})$
- $\text{hasMother} \equiv \text{hasChild}^{-1}$
- $\text{Minni hasMother Mary} \rightarrow \text{hasMother}(\text{Minni}, \text{Mary})$
- $\text{Minni} \neq \text{Missy}$



Ex: ..axioms

- 1. $\text{Child} \sqsubseteq \text{Person}$
- 2. $\exists \text{hasChild}.\top \sqsubseteq \text{Mother}$ (domain)
- 3. $\top \sqsubseteq \forall \text{hasChild}.\text{Child}$ (range)
- 4. $\text{Func}(\text{hasChild})$ (functional)
- 5. $\text{InvFunc}(\text{hasChild})$ (inverse-functional)
- 6. $\text{hasChild}(\text{Mary}, \text{Missy})$
- 7. $\text{hasMother}(\text{Minni}, \text{Mary})$
- 8. $\text{Minni} \neq \text{Missy}$

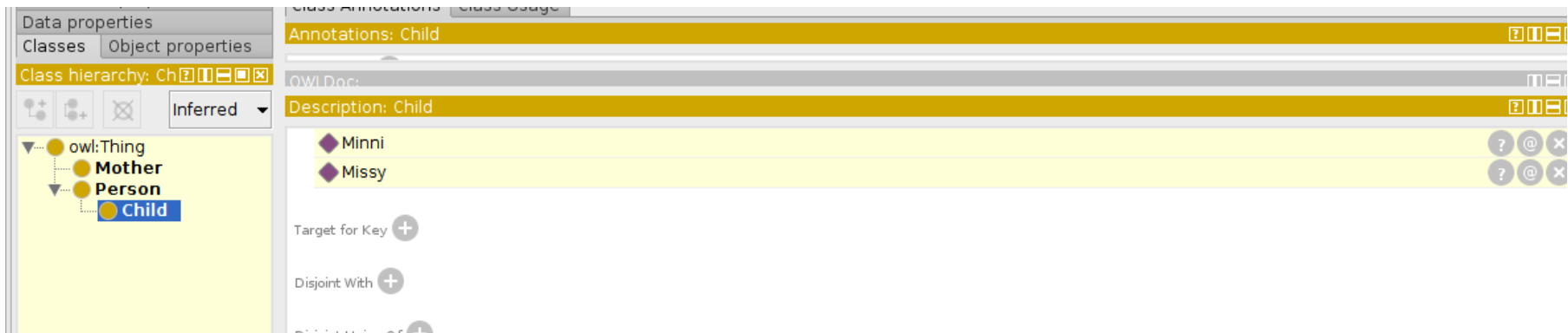


Ex: reasoner infers

- $Mary \in \text{Mother} \rightarrow$ memiliki hasChild, domain adalah Mother
- $Missy \in \text{Child} \rightarrow$ Mary hasChild Missy dan range-nya adalah Child
- $Mary \in \text{Person} \rightarrow \text{Child} \sqsubseteq \text{Person}$, juga $Missy \in \text{Child}$
- $Missy \neq \text{Minni} \rightarrow ?$



- Hapus Missy $\neq$ Minni $\rightarrow$?



- Hilangkan karakteristik Func(hasChild) $\rightarrow$?

